Clinical ICU Patient Phenotyping & Sepsis Severity Profiling

Project Report | Python, scikit-learn, pandas, NumPy, SciPy, Matplotlib & Seaborn

1. Objective

The objective of this project is to explore heterogeneous intensive-care patient records and identify groups of patients with similar clinical profiles, while flagging observations that are unusually unlikely under the learned patient-profile model. The pipeline combines data cleaning, dimensionality reduction, unsupervised phenotyping, anomaly scoring, and cluster-specific regularized regression to help investigate patterns in ICU data and potential markers associated with a length-of-stay/severity measure.

This is an exploratory data-analysis and machine-learning project. Its outputs are not a validated sepsis diagnostic system, a treatment recommendation, or a substitute for clinical judgement.

2. Problem Being Addressed

ICU datasets contain many measurements collected over time, with substantial missingness and correlated clinical variables. Analysing every variable directly can be difficult, and a single average patient profile may conceal meaningful differences between patient subgroups. The project addresses this analytical challenge by compressing the feature space, grouping records/patient summaries into data-driven phenotypes, and identifying low-density observations for further review.

3. Dataset Overview

Item	Observed in supplied CSV
Rows × columns	1,552,210 × 44
Unique patient IDs	40,336
Hourly rows with SepsisLabel = 1	27,916
Patients with at least one SepsisLabel = 1 row	2,932
Mean rows per patient	38.5
Maximum rows for one patient	336

The file contains hourly clinical variables such as heart rate, oxygen saturation, temperature, blood pressure, respiratory rate, laboratory measurements, age, ICU length of stay (ICULOS), SepsisLabel, and Patient_ID. Rows are repeated measurements, not independent patients.

Examples of missingness in the raw data

Column	Missing values
Bilirubin_direct	99.8%
Fibrinogen	99.3%
TroponinI	99.0%

Column	Missing values
Bilirubin_total	98.5%
Alkalinephos	98.4%
AST	98.4%

4. Workflow and Methods

4.1 Data ingestion and patient-level aggregation

The script loads the CSV and detects a patient identifier. When repeated records are present, it aggregates numeric measurements by patient using mean, maximum, and minimum statistics. This converts longitudinal rows into a patient-summary table before modelling. The script then searches for a target column, prioritising ICULOS/ICULOS_max before SepsisLabel, and removes selected identifier/metadata columns.

4.2 Missing-data handling and transformation

Numeric features with at least 80% missingness are excluded. Remaining missing values are imputed with each feature's median. A Yeo–Johnson PowerTransformer is then fitted with standardisation to reduce skewness and put features on a comparable scale. This transformation can handle zero and negative values.

4.3 Dimensionality reduction: PCA and Truncated SVD

PCA is fitted to the transformed matrix, and the number of components is selected to reach at least 85% cumulative explained variance. The code extracts PC1/PC2 loadings and saves a plot of cumulative variance and the largest absolute PC1 loadings. Truncated SVD is mentioned in the script header but is not actually run in the current pipeline.

4.4 Unsupervised patient phenotyping

For candidate cluster counts K = 2 through 7, the script fits K-Means and Gaussian Mixture Models (GMM). It compares silhouette scores for both approaches and AIC/BIC for GMM. The final GMM cluster count is selected by minimum BIC, and each row receives a hard cluster assignment plus posterior membership probabilities.

4.5 Low-density anomaly profiling

The fitted GMM scores each observation using log density. Observations below the 2nd percentile of these scores are flagged as anomalies. This is a statistical outlier flag, not a confirmed diagnosis of shock or sepsis severity.

4.6 t-SNE visualisation

The script projects PCA representations into two dimensions with t-SNE for visual inspection. If there are more than 4,000 observations, it takes a reproducible random sample of 4,000. The plot colours points by GMM phenotype and outlines flagged anomalies.

4.7 Cluster-stratified Lasso and Ridge regression

Within each cluster containing at least 30 observations, the code makes a 75%/25% train-test split and uses cross-validation over 50 alpha values to fit Lasso (L1) and Ridge (L2) regression. It reports test-set R^2, selected alpha, and the number of nonzero Lasso coefficients, then plots selected coefficient patterns across clusters.

5. Outputs Produced by the Script

File	Purpose
plot_1_pca_variance_and_loadings.png	Cumulative explained variance and top PC1 loadings
plot_2_clustering_metrics.png	GMM AIC/BIC and K-Means/GMM silhouette comparison
plot_3_tsne_phenotypes_anomalies.png	2D t-SNE view of phenotypes and flagged outliers
plot_4_cluster_biomarkers.png	Lasso coefficient patterns across clusters

6. Findings Supported by the Supplied Data and Code

The supplied CSV has 1,552,210 hourly rows and 40,336 unique patient IDs. There are 27,916 rows labelled SepsisLabel = 1, across 2,932 patients who have at least one positive label. The average patient contributes 38.5 rows, with a maximum of 336 rows.

The code is designed to generate data-driven phenotypes and low-density outlier flags, and to inspect how regularized regression coefficients vary by phenotype. The actual selected cluster count, anomaly count, PCA component count, test R^2 values, and influential features depend on executing the script. They are intentionally not asserted here because no execution output was supplied as evidence for those results.

7. Important Implementation Notes and Limitations

Potential target leakage: after aggregation, the code selects ICULOS_max as the regression target when available, but SepsisLabel_max may remain among the predictors. If the goal is to predict length of stay, including a sepsis label derived from the same stay may introduce information leakage or confounding. Review the feature set and remove any variables unavailable at prediction time.

Target fallback: if no target is detected, the script generates a synthetic target from transformed features. In that case, regression R^2 measures recovery of an artificial formula, not clinical prediction performance. The report should disclose if this fallback was used.

Preprocessing leakage: imputation, power transformation, and PCA are fitted before the train-test split used in downstream regression. For a rigorous predictive evaluation, fit preprocessing only on training data inside a pipeline, then transform the test set.

Patient-level validation: the current script aggregates by patient first, which reduces repeated-row leakage, but the regression split is performed separately within each cluster rather than using a held-out external cohort. Performance estimates may not generalise to another hospital or population.

Clinical interpretation: unsupervised clusters are mathematical groupings and need clinical review, stability analysis, and external validation before being described as clinical phenotypes. Low-density observations should be called statistical anomalies unless independently confirmed.

Code/report mismatch: Truncated SVD is named in the header but not used; t-SNE is a visualisation step and should not be treated as a predictive model. The script uses the raw SepsisLabel values as binary labels only if selected as the target, but the current target search checks ICULOS first.

8. How to Run

Because the CSV is too large to store in the GitHub repository, it is hosted externally.
Open dataset/README.md in the repository and follow the dataset download link and instructions.
Download Sepsis_data.csv and place it in the same directory as Code.py. Alternatively, pass its path using --csv.
Install the required packages, then run:
python Code.py --csv Sepsis_data.csv --id_col Patient_ID --target_col ICULOS
The script writes four PNG figures to the current working directory. It does not export cluster assignments,
anomaly flags, or regression metrics to a CSV file.

9. Conclusion

This project provides an end-to-end exploratory pipeline for summarising longitudinal ICU records, reducing clinical feature dimensionality, discovering data-driven patient groups, highlighting observations in the low-density tail, and examining cluster-specific regularized associations with a selected target. Its main value is structured exploration of complex clinical data. Results should be treated as hypothesis-generating until the implementation is checked for leakage, evaluated with a sound validation design, and reviewed by clinical experts.

Reproducibility note: dataset statistics in this report were calculated directly from the supplied Sepsis_data.csv. Method descriptions were derived from the supplied Code.py. Model-specific results are not invented.